Unit 4: Concept of Cyber Security

4.1 Cyber Security

Definition:

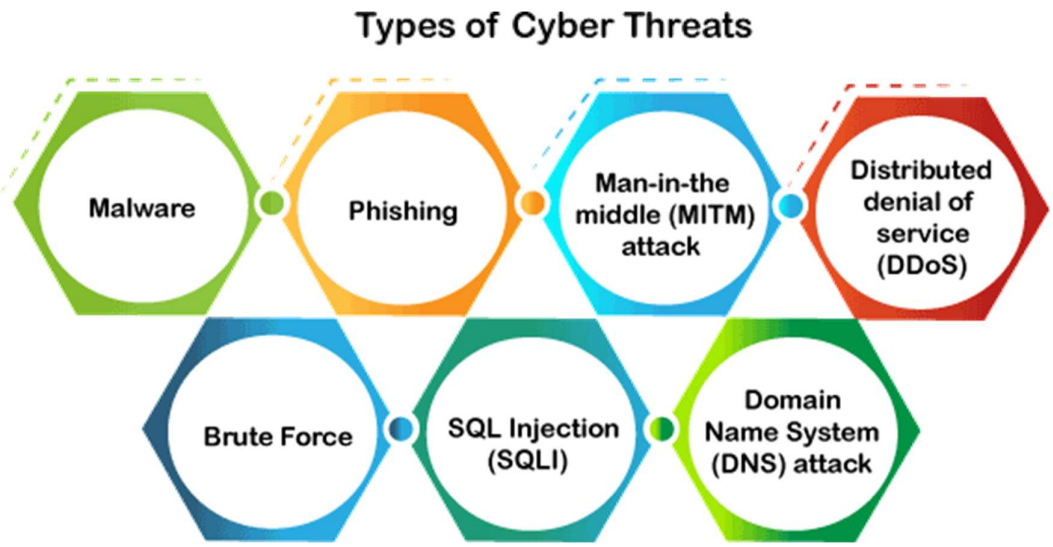
Cybersecurity refers to the practice of protecting internet-connected systems like computers, servers, mobile devices, electronic systems, networks, and data from Harmful attacks.

- It consists of two words:

1. **Cyber**: Refers to technology, including systems, networks, programs, and data.
2. **Security**: Involves the protection of these systems, networks, applications, and information.

4.1.1 Types of Cyber Threats

A cyber threat is any Harmful activity by an individual or group that aims to steal data, gain unauthorized access to a network, or disturb digital systems.



1. Malware (Harmful Software)

- **What is it?**
- Malware is any harmful software created to damage, disturb, or steal information from the user's computer or network.
- **How does it work?**
- Malware can infect the user's system when files are downloaded, malicious websites are visited, or infected devices are plugged in.
- **Examples of Malware:**
  - **Virus:** Attaches to files and spreads when the user opens them. *Example:* A Word file with a virus that corrupts other files.
  - **Worms:** Spread automatically across networks without user action. *Example:* A worm slowing down internet speed by copying itself repeatedly.
  - **Trojan horse:** Pretends to be a useful program but harms the system. *Example:* A game app that secretly steals the user's personal data.
  - **Ransomware:** Locks the user's data and asks for payment. *Example:* Hackers encrypt files and demand money to unlock them.

2. Phishing

- **What is it?**
- Phishing tricks the user into sharing sensitive information like passwords or bank details.
- **How does it work?**
- Attackers send fake emails or messages pretending to be from trusted sources like banks or companies.
- **Example:**

  An email is received saying the user's bank account will be blocked unless a link is clicked and logged into. When this is done, attackers steal login details.

3. Man-in-the-Middle (MITM) Attack

- **What is it?**
- An attacker secretly listens to or alters communication between two people or systems.
- **How does it work?**
- The attacker places themselves between the sender and receiver, pretending to be part of the communication.
- **Example:**

  While using public Wi-Fi at a coffee shop, an attacker intercepts messages and steals the user's login credentials while accessing their bank account.

4. Distributed Denial of Service (DDoS) Attack

- **What is it?**
- A DDoS attack overloads a website or system with so much traffic that it crashes and becomes unavailable.
- **How does it work?**
- Hackers use multiple devices or servers to send fake requests, overwhelming the target system.
- **Example:**

  A popular online store gets so much fake traffic during a sale that real customers cannot access it.

5. Brute Force Attack

- **What is it?**
- A method where attackers guess passwords by trying every possible combination.
- **How does it work?**
- Automated tools are used to test millions of password combinations quickly.

- **Example:**

  A hacker uses a program to guess a weak password like “12345” or “password123.”

6. SQL Injection (SQLI)

- **What is it?**
- An attack that targets databases by inserting harmful code into them.
- **How does it work?**
- Hackers find weaknesses in a website’s code to steal or manipulate data stored in its database.
- **Example:**

  A hacker enters malicious code into a login form, gaining access to all user accounts and details stored on the site.

7. Domain Name System (DNS) Attack

- **What is it?**
- An attack on the DNS, which works like the internet’s phonebook, to redirect users to fake websites or disrupt services.
- **How does it work?**
- Hackers modify DNS records so users trying to visit a authentic website are redirected to a fake one.
- **Example:**

  The user types "amazon.com," but due to a DNS attack, they are sent to a fake website that looks like Amazon and steals credit card details.

4.1.2 Advantages of Cybersecurity

Cybersecurity focuses on protecting systems, networks, and programs from digital attacks, which often aim to:

- Access or steal sensitive information.

- Blackmail users for money.
- Interrupt normal processes.

Advantages of Cybersecurity:

1. **Protects businesses from cyberattacks and data breaches**
   - Prevents financial losses and reputational damage.
2. **Secures data and networks**
   - Ensures that sensitive data is protected from theft or manipulation.
3. **Prevents unauthorized access**
   - Blocks hackers or malicious actors from gaining control of systems or accounts.
4. **Allows quicker recovery after an attack**
   - Reduces downtime and ensures faster recovery of services.
5. **Protects end-user devices like computers and mobile phones**
   - Safeguards devices against malware, phishing, and other threats.
6. **Ensures internet safety by monitoring incoming and outgoing data**
   - Identifies and stops suspicious activity to maintain secure browsing.
7. **Supports business continuity**
   - Keeps systems running smoothly during attacks, reducing disturbances.
8. **Builds trust among developers, stakeholders, and consumers**
   - Display a commitment to security, increasing confidence in products and services.
9. **Provides privacy and security for personal information**
   - Protects sensitive data like passwords, financial details, and personal records.

4.2 Basic Terminology

1. IP Address (Internet Protocol Address)

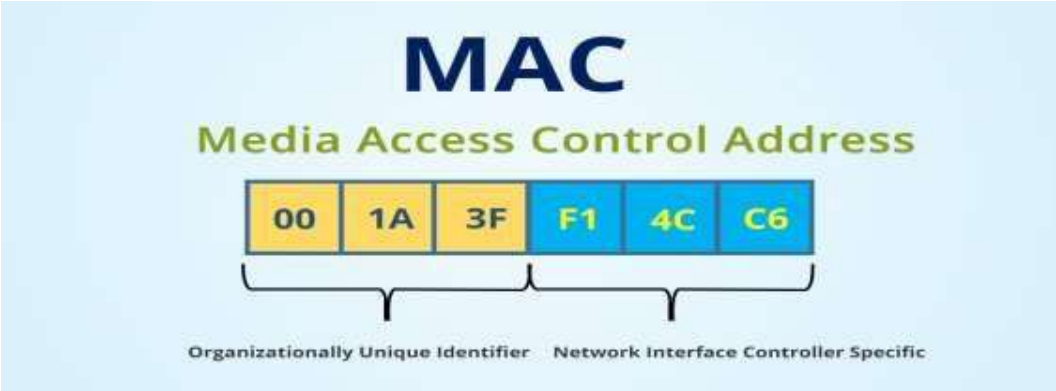
- **What is it?**
- An IP address is a unique number that identifies a device (like a computer, phone, or printer) on a network.
- **How does it work?**
- It works like your home address. It tells the internet where to send data.

- **Example:**
  - A common IP address looks like this: 192.168.1.1. (Four numbers separated by periods, each ranging from 0 to 255).
- **Uses:**

1. Locates devices on a network.
   - Example: Your phone's IP address helps it connect to the Wi-Fi network.
2. Identifies where data needs to go.
   - Example: When you load a website, the server knows your IP to send the page.

2. MAC Address (Media Access Control Address)

- **What is it?**
- A MAC address is a unique code given to your network device (like Wi-Fi or Bluetooth).
- **How does it work?**
- It is like a device's fingerprint, helping it be recognized on a network.



- **Example:**
  - A MAC address looks like this: 00:1A:2B:3C:4D:5E.
  - MM:MM:MM:SS:SS:SS (6-byte hexadecimal format).
- **Uses:**

1. Identifies devices on a network.
   - Example: A router uses MAC addresses to know which devices can connect to the internet.
2. Controls network access.
   - Example: Schools can block unknown devices using their MAC addresses.

3. DNS (Domain Name System)

- **What is it?**
- DNS is like the internet's phonebook. It translates website names into numbers (IP addresses) so computers can understand them.

- **How does it work?**
- When users type a website name (like google.com), the DNS finds it is matching IP address (like 142.250.190.14) and connects the browser to the website.
- **Example:**
  - Instead of remembering a number like 192.0.2.44, users can simply type amazon.com.
- **Uses:**

1. Makes internet browsing easier.
   - Example: You type facebook.com instead of its IP address.
2. Helps computers find websites.
   - Example: DNS directs your browser to the right server when you search online.

Key Differences

Term	What it Does	Example
IP Address	Identifies a device on a network	192.168.1.1
MAC Address	Identifies a network device	00:1A:2B:3C:4D:5E
DNS	Translates website names	google.com → 142.250.190.14

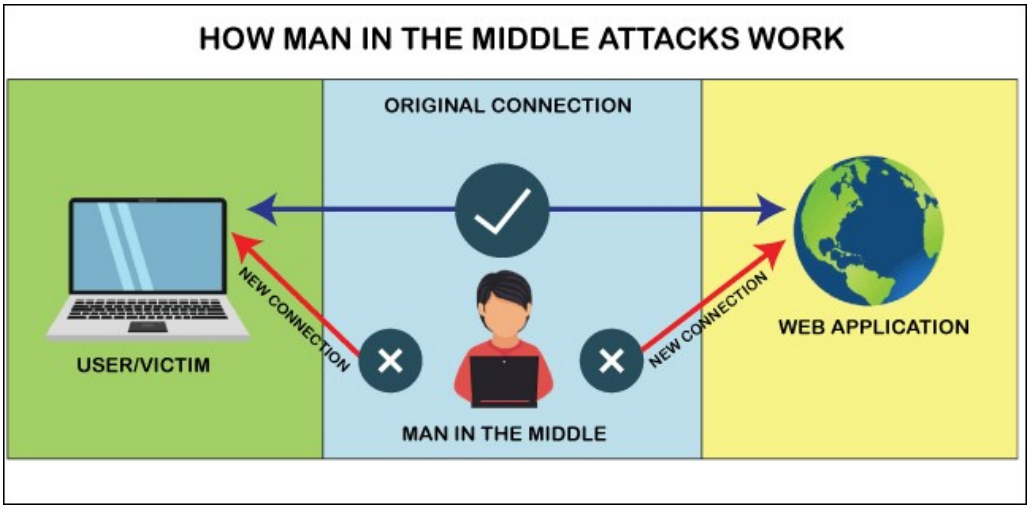
4.3 Types of Attacks

Man-in-the-Middle (MITM)

- **What is it?**

  A Man-in-the-Middle (MITM) attack is a cyber-attack where a hacker secretly places themselves between two parties (like a user and a website) to steal or change information being shared.

- **How does it work?**
  - The hacker makes both parties believe they are safely communicating with each other.
  - However, the hacker is in the middle, reading or altering the data.



- **Example:**
  - Suppose a user tries to log in to an online bank account. Instead of connecting to the bank's real website, the user unknowingly connects to a fake website created by the hacker.
  - The fake website collects the user's login details and passes them to the real bank's website so the user does not notice anything wrong.
- **Why hackers use it:**
  1. To steal sensitive information like credit card numbers or passwords.
  2. To spy on private conversations or meetings for gaining business secrets.

Malware (Malicious Software)

- **What is it?**
  Malware is any harmful software that is created to damage, steal, or gain unauthorized access to a user's computer, phone, or network without their permission.

- **What does it do?**
  - Malware can steal data, damage systems, or allow hackers to control a device.
  - It can affect computers, mobile phones, servers, and even entire networks.

➢ ***Types of Malware (with Examples):***

1. Viruses

- **What is it?**

  A virus is a harmful program that attaches itself to files or programs. When the infected file is opened or shared, the virus spreads to other systems.
- **How it works:**
  - A user opens a file containing the virus.
  - The virus gets activated and spreads to other files or programs on the same device.
- **Example:**
  - A virus can delete important documents on the user's computer or make the computer run very slowly.

2. Worms

- **What is it?**

  A worm is like a virus but can spread itself without needing a host program or file. Worms spread through networks and shared folders.
- **How it works:**
  - A worm makes copies of itself and spreads to other computers.
  - Worms slow down networks by overloading them with data.
- **Example:**
  - A worm spreads through a company's shared network and slows down all employees' systems.

3. Spyware

- **What is it?**

  Spyware is a program that secretly collects private information (like passwords or browsing history) from a user's device.
- **How it works:**
  - It runs in the background without the user knowing and sends information to the hacker.
- **Example:**
  - A spyware program might collect a user's credit card details while they shop online and send the details to the hacker.

4. Trojan Horse

- **What is it?**

  A Trojan horse is malware disguised as a useful or safe file, such as a game or an image.
- **How it works:**

- When a user downloads and opens the Trojan horse file, it secretly performs harmful actions like deleting files or giving hackers access to the user's system.

- **Example:**
  - A free game file downloaded from an untrusted website may look safe but might be a Trojan horse that steals passwords.

5. Logic Bombs

- **What is it?**

  A Logic Bomb is a harmful program that remains inactive until triggered by an event, such as a specific date or time.

- **How it works:**
  - Once triggered, it performs harmful tasks like deleting data or damaging hardware.

- **Example:**
  - A Logic Bomb may be programmed to delete all company data on a specific date, causing major disruption.

6. Ransomware

- **What is it?**

  Ransomware locks or encrypts the user's computer or files, making them unusable until the user pays money (a ransom) to the hacker.

- **How it works:**
  - The hacker encrypts the data with a secret key.
  - The user is asked to pay the hacker to get the key and recover the data.

- **Example:**
  - A company's files are encrypted by ransomware, and the hacker demands payment to unlock them.

7. Keyloggers

- **What is it?**

  A Keylogger records everything the user types on their keyboard and sends the information to the hacker.

- **How it works:**
  - The Keylogger tracks usernames, passwords, and other sensitive information typed by the user.
- **Example:**
  - A hacker installs a Keylogger on a public computer to collect users' email passwords.

✸ DDoS (Distributed Denial of Service)

What is a DDoS Attack?

- A **DDoS attack** happens when many computers send too much traffic to one system, like a website, causing it to stop working.
- Hackers using viruses like Trojans usually control the attacking computers.

How does it work?

- Hackers send so many requests to the system that it cannot handle them all.
- This overload makes the system slow or crash, so it cannot serve real users.

Example:

- Think of a shopping mall with too many people rushing in at once. The doors are blocked, and no one can go inside.

Goal of a DDoS Attack:

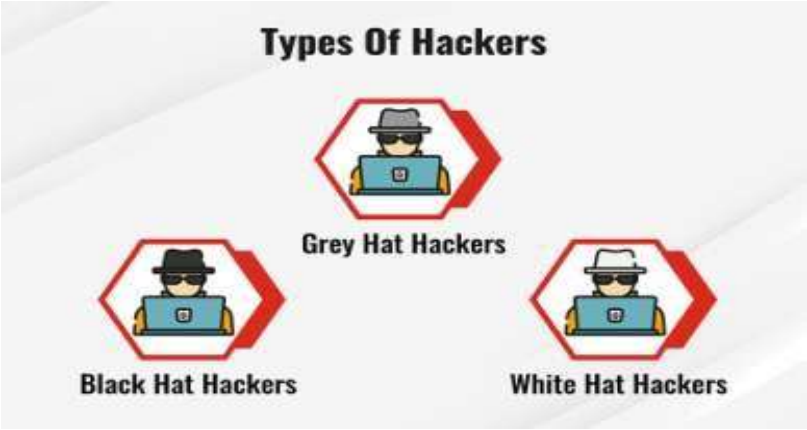
- The main goal is to **block the target system** from working properly by exhausting its resources.
- These attacks often target websites or servers to make them unavailable.

✸ Hackers

Who are Hackers?

- Hackers are skilled computer experts who can enter systems by bypassing security.
- Some hackers are **bad (Black Hat)**, and some are **good (White Hat)**.

✦ Types of Hackers



1. Black Hat Hackers (Bad Hackers)

- **Who are they?**
  - Black hat hackers are **criminals** who break into systems to steal data, cause damage, or earn money illegally.
- **What do they do?**
  - Plant viruses like Trojans and ransomware.
  - Steal sensitive information like passwords or bank details.
  - Attack websites and servers (e.g., DDoS attacks).
- **Example:**
  - A black hat hacker may hack into a bank's system to steal money or customer information.

2. White Hat Hackers (Good Hackers)

- **Who are they?**
  - White hat hackers are **ethical hackers** who help companies protect their systems from attacks.
  - They follow rules and work for organizations or governments.
- **What do they do?**
  - Find and fix weak points in systems.
  - Test systems to ensure they are safe from cyberattacks.
  - Protect personal and business data.
- **Example:**
  - A white hat hacker may be hired by a company to test its security and prevent cybercriminals from attacking.

3. Gray Hat Hackers

- Gray hat hackers are in between black hat and white hat hackers**.**
- They do not hack for bad reasons (like black hats) but also do not always follow rules (like white hats)**.**

What do they do?

- They may find security weaknesses in a system without permission and then inform the owner.
- Sometimes, they ask for money to fix the issue they found.

Key Points:

- They **do not harm systems** like black hats.
- They **do not follow ethical guidelines** like white hats.
- Their goal is usually to **show off skills or help improve security**.

Example:

- A gray hat hacker finds a security issue in a company's website **without permission** and informs the company.
- They might say, "I found this problem; if you pay me, I can fix it for you."
- The company did not ask for their help, so their actions are **not legal**, but they **do not harm the system intentionally**.

Difference between Black, White, and Gray Hat Hackers

Type of Hacker	Follows Rules?	Purpose	Example Action
Black Hat	No	Harm systems, steal data, or earn money illegally.	Hacks a bank to steal money.
White Hat	Yes	Protect systems and find weaknesses ethically.	Tests a company's security with permission.
Gray Hat	Sometimes	Improve security but without permission.	Finds a website issue without asking, and then informs the owner.